

HW

- 1. What is the time complexity of removing an element from the end of an array and why?**
- 2. Consider the array [8, 3, 12]. What is the time complexity of adding the number 5 at the second position, and why?**
- 3. If you want to remove the first element in the array [8, 3, 12], what would be the time complexity and the reason for this complexity?**
- 4. What would be the time complexity of inserting an element at the third position of the array [8, 3, 12] if the array is not full?**

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4. Given the pseudocode below, what is the time complexity of adding a new element at the beginning of an array, and why?

```
function addAtBeginning(array, new_element)  
  for i from length(array) to 1 do  
    array[i] = array[i-1]  // Shift all elements to the right  
  end for  
  array[0] = new_element  // Insert new element at index 0  
end function
```

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5. Examine the pseudocode below. What is the time complexity of removing an element from the end of an array and why?

```
function removeAtEnd(array)
    last_element = array[length(array) - 1] // Access the last element
    array[length(array) - 1] = null        // Remove the last element
    return last_element
end function
```

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6. In the provided pseudocode, how is the 5th element of the array accessed in constant time ($O(1)$) using the formula $\text{array_addr} + \text{elem_size} \times (\text{index} - \text{first_index})$?

```
function accessElement(array_addr, elem_size, index, first_index)
    offset = elem_size * (index - first_index)
    element_address = array_addr + offset
    return element_at(element_address)
end function
```

```
array_addr = 1000    // Base address of the array
elem_size = 4        // Each element is 4 bytes
index = 5            // Access the 5th element
first_index = 0      // Arrays are 0-indexed
```

```
element = accessElement(array_addr, elem_size, index, first_index)
```